

Skip One Layer? A Multilingual and Multi-Task Study of Single-Layer Skipping in Modern Language Models

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Abstract

Modern language models are typically deployed using all transformer layers during inference, implicitly assuming that every layer contributes positively to downstream performance. Prior studies on layer skipping often focus on limited settings and commonly associate removable layers with specific transformer regions, particularly middle layers. However, whether single-layer skipping generalizes across diverse tasks, languages, and model architectures remains underexplored.

In this work, we conduct a multilingual and multi-task empirical study of single-layer skipping in modern instruct-tuned language models. We evaluate 9 text-only models spanning multiple model families and parameter scales, including Gemma 3, Llama 3.2, and Qwen 3, OLMo 2 without modifying model parameters. Our experiments systematically evaluate single-layer skipping across all layers on diverse benchmarks covering NLU, reasoning, bias evaluation, code generation, instruction following, and multilingual knowledge evaluation, including MMLU across 57 subject domains and 10 typologically diverse languages.

Our results show that the effect of single-layer skipping is highly context-dependent. We show that skipping a single transformer layer is an effective strategy for leveraging the capabilities of a language model: there is a high chance of achieving better benchmark performance while simultaneously reducing computation during inference. However, the optimal skipped layer varies depending on model architecture, task, domain, and language. Beneficial skipped layers occur most frequently in the second half of the network, though they also emerge in early layers in some cases. We further observe that performance improvements are more common in structured prediction tasks, whereas degradations more frequently occur in tasks requiring precise generation.

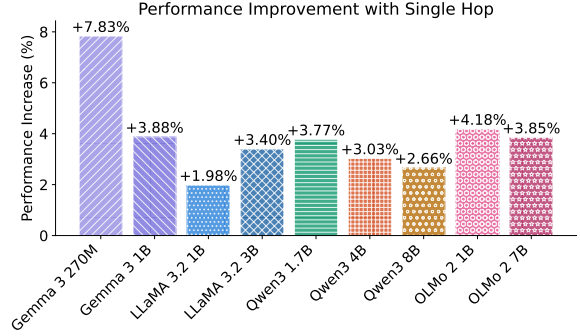


Figure 1: **Performance improvement (averages) from single-layer skipping on benchmarks.** Beyond reducing computation, this approach yields a median benchmark gain of 2.80% on above models, with peak improvements exceeding 30% on select reasoning tasks and over 5% on lower-resource languages.

1 Introduction

Language models have demonstrated strong capabilities across a wide range of downstream tasks, including question answering, reasoning, translation, and domain-specific applications. Over time, these models have continuously grown in scale, with many studies showing that increasing the number of parameters often leads to improved benchmark performance (Zhao et al., 2026; Ignat et al., 2024). Consequently, model scaling has become a dominant direction in language model development, with many recent models containing hundreds of billions of parameters (Xiao et al., 2025; Mienye et al., 2025). Yet, expanding parameter counts no longer guarantees proportional performance gains, as substantial increases in model size sometimes produce only marginal improvements on benchmark results (Zhao et al., 2026; Cheng et al., 2024). This raises an important question regarding whether current language models utilize their computational resources in the most efficient manner for specific downstream tasks.

Current efforts to enhance language model per-

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formance have primarily focused on improving model capabilities through various weight modification and optimization strategies. Common approaches include instruction tuning (Ma et al., 2026; Zhang et al., 2024), fine tuning (Wang et al., 2025; Jo et al., 2025), and neuron-level interventions to enhance task-specific capabilities and model behavior (Huang et al., 2025; Antverg et al., 2022). While these methods have demonstrated effectiveness, they mainly aim to increase model intelligence and task performance rather than investigate whether the inference process itself can be executed more efficiently. Comparatively fewer studies have examined the possibility of improving inference efficiency without modifying the underlying model parameters (Leviathan et al., 2023; Dao et al., 2022).

This study investigates whether language models can achieve more efficient inference without modifying the underlying weights. This weight-preserving approach is particularly relevant for domain-specific applications, where downstream tasks and input characteristics are already predefined. In many practical scenarios, modifying model weights through fine-tuning is not feasible due to computational constraints, deployment limitations, maintenance costs, or the need to preserve a shared model across multiple applications (Hacker and Holweg, 2026; Wang et al., 2025). This study conducts a systematic investigation into the layer-skipping phenomenon in modern language models. We explore whether selectively bypassing a single intermediate layer during the standard forward pass can maintain or even improve downstream performance without modifying any model weights. To deeply understand this phenomenon, we systematically evaluate single-layer skipping across multiple language model architectures, diverse downstream benchmark tasks, and various input languages to examine how inference behavior varies depending on these contexts.

Although prior work has shown that transformer layers may exhibit redundancy, the generality of single-layer skipping remains insufficiently understood. Existing studies often evaluate layer removal or skipping under restricted settings, such as a limited set of tasks, a single model family, monolingual benchmarks, or efficiency mechanisms that require additional routing modules, calibration data, or optimization procedures. Moreover, several prior analyses implicitly or explicitly associate skippable layers with particular depth re-

gions, especially middle layers. These limitations make it unclear whether single-layer skipping is a general inference-time phenomenon or a context-dependent behavior shaped by model architecture, task type, domain, and input language.

To address this gap, this work conducts a systematic empirical study of single-layer skipping across multiple instruct-tuned language model families, parameter scales, downstream tasks, input languages, and MMLU subject domains. Rather than proposing a trainable routing or modifying model parameters, we evaluate a simple weight-preserving intervention in which one transformer layer is bypassed during inference. This allows us to empirically characterize single-layer skipping across the evaluated settings, including cases where it improves performance, cases where it degrades performance, and whether beneficial skipped layers remain confined to a fixed depth region or vary across model, task, domain, and language contexts.

The main contributions are as follows:

- We conduct a systematic empirical study of single-layer skipping in modern instruct-tuned language models, evaluating whether bypassing one transformer layer at inference time can maintain or improve downstream performance without modifying model parameters.
- We extend prior layer-skipping and layer-removal analyses beyond narrow evaluation settings by considering multiple model families, parameter scales, downstream tasks, input languages, and MMLU subject domains.
- We show that performance-improving skipped layers exist across models (Figure 1) and are highly context-dependent. Although beneficial skips mostly occur in the second half of the network, they are not confined to a universal depth region and can sometimes appear in earlier layers depending on the model, task, domain, and language.
- We identify both positive and negative cases of single-layer skipping, showing that the method can improve performance in many multilingual and reasoning settings while degrading performance on tasks requiring precise generation (e.g. arithmetic, symbolic reasoning, or sequential state tracking).

This study provides foundational insights for improving inference efficiency in applied AI systems

utilizing language models, particularly for domain-specific applications where task requirements and input languages are already well defined. By exploring efficient inference pathways without requiring model retraining or parameter modification, this work highlights new opportunities for practical deployment of language models with reduced computational cost.

2 Related Works

The computational cost of deploying large language models (LLMs) has motivated extensive research into inference efficiency, primarily through *model compression* and *structural reduction*. Compression methods like quantization (Zandieh et al., 2026; Lin et al., 2025) and structured pruning (Li et al., 2026; Ling et al., 2024) effectively decrease parameter counts before deployment but typically require additional training or optimization to maintain performance (Zhu et al., 2024; Ma et al., 2023). Architectural redundancy approaches permanently remove intermediate layers. For instance, Liu et al. (2025) demonstrates that many layers produce highly similar representations, enabling compression through low rank adaptive parameters, while task aware pruning (Naim et al., 2026) can simultaneously improve accuracy and reduce latency. Dynamic methods offer an alternative by adapting inference at runtime. For example, early exiting halts computation once a confidence threshold is reached (Chen et al., 2024; Xin et al., 2020; Schwartz et al., 2020), while layer skipping methods such as Had-Skip (Wang et al., 2023), D³ (Fan et al., 2025), and DiffSkip (Luo et al., 2025) selectively replace layers or sublayers based on input characteristics. However, these approaches often still require trainable components, optimization procedures, or calibration data.

Beyond these methods, recent studies reveal substantial flexibility in transformer internal structures. Model stitching research shows that intermediate representations across independently trained models can be connected with lightweight transformation, implying representational compatibility and redundancy within transformer architectures (Bansal et al., 2021). Inference time interventions further demonstrate that model behavior can be steered by manipulating hidden representations at specific layers without any parameter updates (Nguyen et al., 2025; Li et al., 2023), and intermediate layers can sometimes encode richer repre-

sentations than final layers for downstream tasks (Skean et al., 2025). Despite these advances, two gaps remain. First, most layer skipping methods are evaluated in narrow settings. Second, the assumption that redundant layers concentrate in middle depth regions has not been thoroughly tested across architectures, tasks, and languages. To address this, we investigate *Single Hop*, a lightweight inference strategy that skips a single intermediate transformer layer without parameter modification, using a simple intervention to empirically characterize single layer skipping across multiple model families, diverse benchmarks, ten languages, and 57 MMLU subject domains.

Recent studies such as ShortGPT (Men et al., 2024) and the Unreasonable Ineffectiveness of Deeper Layers (Gromov et al., 2024) demonstrate that LLMs exhibit layer redundancy, proposing the outright truncation of final layers. However, these approaches operate under a depth-localization assumption, presuming that redundancy is heavily concentrated in specific areas. This overlooks the highly context-dependent nature of intermediate computations.

On the other hand, methods like TELL-TALE (Naim et al., 2026) establish that layer pruning must be task-aware to prevent capability degradation. Furthermore, existing literature primarily treats redundancy as a monolingual phenomenon. Our results extend the task-aware paradigm by revealing that layer utility is also fundamentally *language-dependent*.

3 Method

3.1 Single Hop

To investigate the layer-skipping phenomenon, we conduct a systematic empirical evaluation of whether individual transformer layers contribute positively or negatively to downstream task performance. Rather than proposing a new inference algorithm or modifying model parameters through additional training, we use single-layer skipping as a simple intervention during inference to analyze how bypassing individual layers affects model behavior across different tasks, languages, and architectures.

For a language model consisting of n layers, we select a layer X to skip during the forward pass. Under the standard inference process, information flows sequentially from layer 1 to layer n . In the *Single Hop*, the output from layer $X - 1$ is passed

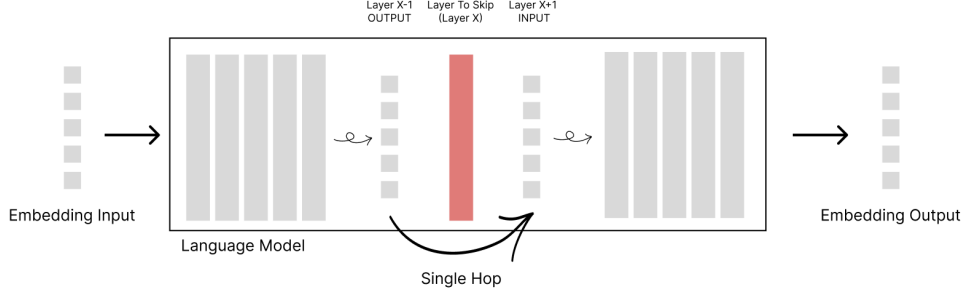


Figure 2: Single Hop.

directly as the input of layer $X + 1$, effectively by-passing the computation of layer X (see Figure 2). Formally,

$$\mathbf{h}_{X+1}^{in} = \mathbf{h}_{X-1}^{out}$$

where $\mathbf{h}_{X-1}^{out}$ denotes the output representation from layer $X - 1$, and $\mathbf{h}_{X+1}^{in}$ denotes the input representation to layer $X + 1$.

No other components of the language model are modified. The remaining layers retain their original processing flow, and inference is conducted using the same procedure as the standard model.

3.2 Benchmarking

We use the LM Evaluation Harness (Gao et al., 2024) framework to assess language model performance on downstream tasks and automate the benchmarking process. Since the original implementation does not provide functionality for skipping layers during inference, we extended the framework by incorporating support for the proposed *Single Hop* mechanism. The implementation details and results are available at github.com/Efficient-Inference-on-Multilingual-LLM/Skip_One_Layer.

3.3 Experiment Setup

For each evaluated language model consisting of n transformer layers, we systematically investigate the effect of skipping individual layers during inference. Specifically, we conduct n separate evaluations, where in each evaluation exactly one layer is skipped, ranging from layer ID 0 to layer ID $n - 1$ (see Algorithm 1).

Let $S(i)$ denote the benchmark performance score obtained when layer i is skipped during inference. The optimal skipped layer is defined as the layer that yields the highest benchmark:

Algorithm 1 Experiment Setup

Require: Language model M with n layers, benchmark dataset D

Ensure: Best option and Optimal skip layer

- 1: Configure inference
 - 2: Evaluate model performance on dataset D
 - 3: Obtain benchmark score $S(base)$, $S(0)$
 - 4: $S_{best_option} \leftarrow S(base)$
 - 5: $L_{best_option} \leftarrow base$
 - 6: $S_{optimal_skip} \leftarrow S(0)$
 - 7: $L_{optimal_skip} \leftarrow 0$
 - 8: **for** $i = 1$ to $n - 1$ **do**
 - 9: Configure inference by skipping layer i
 - 10: Evaluate model performance on dataset D
 - 11: Obtain benchmark score $S(i)$
 - 12: **if** $S(i) > S_{best_option}$ **then**
 - 13: $S_{best_option} \leftarrow S(i)$
 - 14: $L_{best_option} \leftarrow i$
 - 15: **end if**
 - 16: **if** $S(i) > S_{optimal_skip}$ **then**
 - 17: $S_{optimal_skip} \leftarrow S(i)$
 - 18: $L_{optimal_skip} \leftarrow i$
 - 19: **end if**
 - 20: **end for**
 - 21: **return** L_{best_option} , S_{best_option} , $L_{optimal_skip}$, $S_{optimal_skip}$
-

$$L_{best} = \arg \max_{i \in \{0, \dots, n-1\}} S(i)$$

where L_{best} represents the selected layer ID for the *Single Hop* configuration.

3.4 Experiment Details

The list of evaluated models is provided in Appendix A, while the list of tasks and languages is provided in Appendix B. All experiments were conducted using a single H100 GPU. All inference settings for the evaluation followed the LM

Benchmark	Gemma 3 270M IT	Gemma 3 1B IT	LLaMA 3.2 1B IT	LLaMA 3.2 3B IT	Qwen 3 1.7B	Qwen 3 4B	Qwen 3 8B	Olmo 2 1B IT	Olmo 2 7B IT	Macro (all models)	Models > 0
GSM8K	-0.61%	-0.45%	-11.22%	-3.41%	-4.62%	-0.53%	-0.99%	-13.87%	-1.52%	-4.14%	0/9
HumanEval Instruct	+0.61%	+2.44%	-19.51%	-2.44%	-7.93%	+30.49%	+2.44%	-14.63%	+0.61%	-0.88%	5/9
IFEval	-3.51%	+2.40%	-9.98%	-3.33%	+2.96%	+4.25%	+17.93%	-2.03%	+1.66%	+1.15%	5/9
SQuADv2	+6.59%	+4.33%	-5.31%	-0.97%	+1.39%	+3.71%	+2.50%	+0.96%	+1.04%	+1.58%	7/9
ToxiGen	+11.28%	+12.02%	+16.06%	+3.83%	+6.17%	+17.66%	+6.38%	+4.36%	+6.91%	+9.41%	9/9
HellaSwag	-0.71%	-0.22%	-1.51%	-0.66%	-1.42%	-0.76%	-0.36%	-2.76%	-0.77%	-1.02%	0/9
MMLU (avg)	+8.07%	+3.80%	+2.17%	+3.07%	+3.38%	+2.75%	+2.42%	+4.15%	+3.66%	+3.72%	9/9
BBH (avg)	+3.21%	+2.71%	-2.81%	+7.96%	+9.68%	+4.10%	+3.78%	+3.94%	+4.95%	+4.17%	8/9
ANLI (avg)	+1.37%	+1.09%	+1.84%	+2.19%	+1.39%	+1.97%	+0.26%	-0.28%	+1.81%	+1.29%	8/9
Macro avg	+2.92%	+3.12%	-3.36%	+0.69%	+1.22%	+7.07%	+3.82%	-2.24%	+2.04%	+1.70%	7/9

Table 1: Each entry is the absolute percentage of performance difference compared to its normal inference flow showing Single Hop help the model to improve performance in most of downstream tasks and works across models. For MMLU, BBH, and ANLI, average values are reported because these benchmarks consist of multiple evaluation tasks or rounds.

Benchmark	Gemma 3 270M IT	Gemma 3 1B IT	LLaMA 3.2 1B IT	LLaMA 3.2 3B IT	Qwen 3 1.7B	Qwen3 4B	Qwen 3 8B	Olmo 2 1B IT	Olmo 2 7B IT
GSM8K	0	0	0	0	0	0	0	0	0
HumanEval Instruct	2	11	0	0	0	23	1	0	2
IFEval	0	1	0	0	1	7	6	0	6
SQuADv2	10	7	0	0	6	5	8	2	12
ToxiGen	16	17	13	11	12	18	13	4	13
HellaSwag	0	0	0	0	0	0	0	0	0
MMLU	8.03	5.18	2.20	5.34	4.78	6.19	6.14	4.02	8.72
BBH	4.14	3.54	1.28	8.86	6.21	4.5	5.04	5	13.93
ANLI	6.67	8.33	4.67	10.5	2.33	7.67	1.33	0	5.67

Table 2: The number of candidate layers that produce improvements over its corresponding normal inference benchmark score. Candidate = 0 indicates that no skipped layer yields performance improvements (i.e., all configurations result in negative values in Table 1). For MMLU, BBH, and ANLI, average values are reported because these benchmarks consist of multiple evaluation tasks or rounds.

Evaluation Harness default settings. No prompt-formatting or decoding option was overridden, so every setting is the harness default for the task, and that default applies identically to the baseline and to all K skipped variants.

4 Results

4.1 Effectiveness

Based on Table 1, our evaluation of single-layer skipping generally shows positive effects across most evaluated tasks. In addition to reducing computational requirements, the approach improves overall benchmark performance by a median of 2.80% compared to standard inference. Furthermore, we observe highly pronounced gains in specific contexts, achieving improvements over 8% on select models (see Table 3) and over 30% on select reasoning and coding tasks. These findings suggest that language models may not require all of their layers to effectively perform specific downstream tasks, as selectively skipping certain layers can preserve or even improve performance. Generally, the Single Hop approach works across models, with a median success rate of 97.1% across the nine mod-

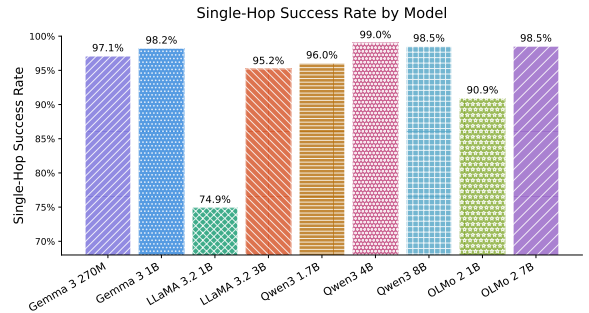


Figure 3: Single-Hop success rate by model. A Single-Hop run is considered successful if the model maintains or improves the baseline performance while bypassing a single layer during inference.

els we evaluated (see Figure 3). This indicates that modern language models generally do not need all of their layers when performing certain specific tasks, highlighting the potential for layer pruning while preserving performance on those tasks.

4.2 Candidates to Skip

In cases where Single Hop produces improvements, we observe that the optimal layer is selected from multiple candidate layers (Table 2). We define can-

Language	Gemma 3 270M IT	Gemma 3 1B IT	LLaMA 3.2 1B IT	LLaMA 3.2 3B IT	Qwen 3 1.7B	Qwen3 4B	Qwen 3 8B	Olmo 2 1B IT	Olmo 2 7B IT
Arabic	+8.68%	+3.48%	+2.77%	+3.20%	+3.03%	+2.51%	+2.64%	+4.50%	+3.78%
English	+8.06%	+3.00%	+0.74%	+2.85%	+3.00%	+2.26%	+1.68%	+3.00%	+2.44%
French	+6.87%	+3.89%	+2.19%	+2.87%	+3.06%	+2.76%	+2.35%	+2.84%	+3.09%
Hindi	+7.91%	+4.22%	+2.85%	+3.33%	+4.17%	+2.97%	+2.60%	+6.09%	+4.98%
Indonesian	+8.12%	+2.87%	+2.38%	+2.84%	+3.23%	+2.26%	+2.47%	+3.71%	+3.84%
Japanese	+7.78%	+3.53%	+2.03%	+2.89%	+3.55%	+2.79%	+2.42%	+5.14%	+4.02%
Korean	+7.86%	+4.43%	+2.60%	+3.12%	+3.19%	+2.36%	+2.23%	+4.29%	+3.78%
Russian	+8.68%	+3.71%	+1.75%	+3.24%	+3.14%	+2.34%	+2.20%	+2.85%	+3.47%
Swahili	+8.28%	+4.92%	+2.61%	+3.57%	+4.98%	+5.42%	+3.52%	+5.18%	+3.64%
Chinese	+8.51%	+3.99%	+1.77%	+2.80%	+2.46%	+1.82%	+2.10%	+3.89%	+3.60%

Table 3: Average absolute percentage difference relative to normal flow, across different input languages, on the MMLU task.

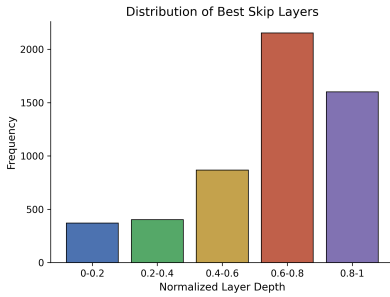


Figure 4: Summary of the Relative Location (Model Depth) of the Best Layer to Skip (i.e., the Layer That Yields the Greatest Benchmark Improvement)

didate layers as the set of layers that yield performance improvements over the baseline benchmark when individually skipped during inference. This suggests that the effectiveness of *Single Hop* is not merely due to chance, as the optimal layer consistently emerges from several beneficial candidates. Furthermore, the presence of multiple candidate layers indicates that performance gains are not tied to one specific layer only, but instead reveal that certain layers may contribute differently depending on the downstream task. This observation suggests that some layers may introduce redundant or less useful processing for particular tasks, and bypassing such layers can improve inference effectiveness. The consistent appearance of multiple beneficial candidates across experiments further supports the idea that layer skipping captures meaningful characteristics of the model’s internal computation rather than random fluctuations in performance.

4.3 Optimal Layer to Skip

Our experiments reveal that the optimal layer to skip varies substantially depending on the language model architecture, downstream task, evaluation

domain, and input language. As shown in Table 2, there is no single universally optimal layer across all evaluated settings, as multiple candidate layers empirically improve performance simply by being skipped. Crucially, the location of performance-improving skipped layers cannot be consistently localized to a fixed depth region (see Figure 4). While we observe that beneficial skip targets occur most frequently in the second half of processing, performance improvements also sometimes emerge when skipping early layers.

This finding demonstrates that layer-skipping behavior is highly context-dependent. The variability indicates that representational redundancy or harmful layers in modern instruct-tuned LLMs emerge from complex interactions among model representations, downstream tasks, and multilingual inference dynamics, rather than being confined to a specific structural block. Consequently, identifying the optimal layer requires a context-aware analysis rather than relying on fixed-region assumptions.

4.4 Multilingual Performance and Analysis

Table 3 demonstrates that *Single Hop* consistently improves performance across all evaluated models and languages. Although the magnitude of improvement varies depending on the language and model architecture, positive gains are observed in nearly all settings.

To better understand these linguistic contexts, we categorize the ten languages using the Joshi taxonomy (Joshi et al., 2020), spanning from Class 1 (lowest resource) to Class 6 (highest). As illustrated in Figure 5, performance gains negatively correlate with resource availability. Lower-resource languages benefit significantly more from *Single Hop*: Swahili (Class 2) sees the highest gains across models (up to 5.42% on Qwen-3-4B), and

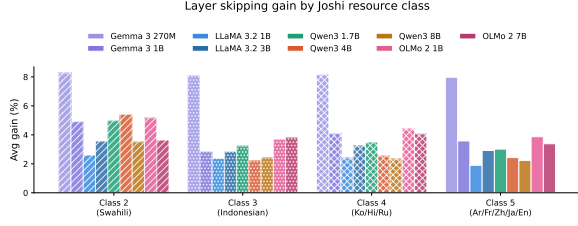


Figure 5: Average layer skipping gain categorized by Joshi resource class. Lower-resource languages (Classes 2–4) consistently exhibit higher performance gains across all evaluated models compared to high-resource languages (Class 5).

Class 3 and 4 languages (Indonesian, Hindi, Russian) consistently outpace Class 5 languages (English, French, Chinese, Arabic, Japanese), which hover between 1.68% and 3.53%.

We find that models efficiently utilize their full depth to process complex reasoning for high-resource languages. Conversely, for lower-resource languages, it is possible that certain intermediate layers act as bottlenecks or introduce noise due to limited training exposure. Bypassing these layers via *Single Hop* may mitigate this interference, though further mechanistic interpretability studies are required to verify these internal dynamics.

Model-Language Affinity and the “Native” Effect Beyond general resource availability, we observe that layer skipping yields the most constrained gains when models process high-resource languages that heavily dominate their pre-training corpora. For example, the English-centric Llama-3.2-1B yields its lowest improvement (0.74%) on English tasks. Similarly, the Chinese-rooted Qwen-3 models generally show notably lower gains for Chinese and English compared to distant languages like Swahili (4.98% and 5.42%) or Hindi (4.17% and 2.97%).

While this pattern is not absolute across every model scale (e.g., Llama-3.2-3B shows a marginally lower gain on Chinese than English, and Qwen-3-8B on English versus Chinese), the broader trend indicates that models possess lower elasticity to layer skipping on their most highly represented languages. We discover that models develop highly optimized, rigid computational pathways for their primary languages, where removing a layer disrupts a finely tuned sequence. For non-primary languages, intermediate layers might perform redundant cross-lingual mapping that is less critical to the final output. However, this insight re-

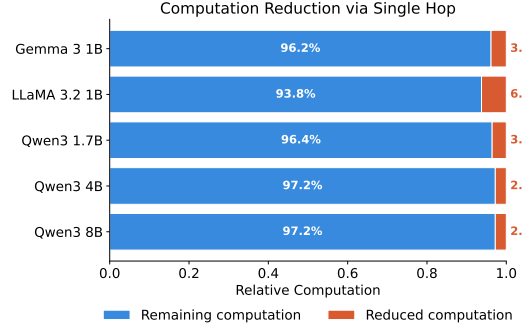


Figure 6: Reduced computation by applying *Single Hop* on each model

quires further discovery, highlighting a promising direction for causal representation analysis.

4.5 Models

The layer-skipping phenomenon is observable across different model architectures. The approach is effective across models from different families with comparable parameter scales. Despite architectural differences among these models, performance improvements are consistently observed. This suggests that the skipping-layer method is not limited to a specific architecture and may be applicable across a broader range of language models.

However, we find that smaller models tend to exhibit more volatile performance under *Single Hop*. In some cases, they achieve substantial improvements, reaching up to twice the baseline performance (e.g., Gemma 3 270M), while in others they suffer severe degradation, dropping to nearly half of the baseline performance (e.g., Llama 3.2 1B Instruct). In contrast, larger models typically achieve more modest improvements of around 3–4% (see Figure 1).

Nevertheless, *Single Hop* remains effective for larger models, as demonstrated by experiments on Qwen-3 models with varying parameter sizes. We observe that larger models tend to exhibit smaller performance gains when a layer is skipped, although positive improvements are still consistently maintained. Furthermore, while larger models tend to achieve smaller average gains, they exhibit a higher success rate for single-layer skipping, i.e., there is almost always some layer that can be skipped with a positive effect. The higher average improvements observed in smaller models are largely driven by a few extreme positive cases, whereas more task instances experience per-

formance degradation relative to the baseline. One possible explanation is that larger models possess greater representational capacity and redundancy, making them more robust to architectural interventions during inference.

From an efficiency perspective (see Figure 6), the relative computational reduction becomes smaller for larger models because a single layer represents a smaller proportion of the overall computation. However, the amount of computation and model parameters bypassed by skipping a layer remains larger in absolute terms, resulting in meaningful reductions in memory usage.

4.6 Statistical Significance

We test whether a *single-layer skip exists that preserves or improves* downstream accuracy. Each evaluation cell yields a matched pair (Baseline, Single Hop), where the Single Hop score is the *maximum* over all K candidate layers. We therefore analyze an **oracle (best-case) selection**: the per-cell delta $\Delta = \max_{k \leq K} s_{\text{skip},k} - s_{\text{base}}$ provides an upper bound on deployable performance. To evaluate these deltas, we apply the **Wilcoxon signed-rank test** (Demšar, 2006) (reporting paired t -test and Cohen’s d_z) with Holm–Bonferroni correction across slices. We also report $p_K = \min(p \cdot K_{\text{slice}}, 1)$; this controls a family of K pre-specified tests and does not remove the optimism of maximizing over those same K layers. Because Δ maximizes over K , we interpret the results strictly as an *existence* proof of beneficial skip layers and treat reported magnitudes as upper bounds. A full track breakdown is provided in Appendix C.

Pooled result. Across $N = 5454$ paired evaluations, a beneficial skip layer is usually found (median $\Delta = +2.80\%$; 90.6% positive). The signed-rank test rejects H_0 ! ; median(Δ) = 0 ($p < 10^{-4}$, $d_z = +0.89$, rank-biserial $r = +0.94$; Table 6). However, given the large N and post-hoc selection inflation, the magnitudes should be read as upper bounds rather than expected gains.

What is robust to the selection. One substantive finding transcend this post-hoc optimism. First, the effect is **strongly capability** dependent: best-case skipping preserves knowledge accuracy (MMLU survives Holm $\circ$ K), but *consistently degrades* exact-reasoning tasks (GSM8K, HellaSwag; rank-biserial -1.0). **In summary, single-layer redundancy is highly task-specific—often benefiting knowledge retrieval, but reliably harming ex-**

act reasoning—and the apparent improvements must be treated strictly as upper bounds rather than expected gains of a fixed rule.

5 Discussion

The proposed Single Hop approach shows that reducing inference computation does not necessarily degrade performance. It often maintains or even improves benchmark performance, suggesting that not all layers contribute equally to downstream performance. While the Single Hop approach is generally effective, the method is still exhaustive and requires substantial resources. Nonetheless, our results motivate exploring more aggressive skipping strategies.

One extension is Double Hop, which skips two consecutive layers; initial experiments suggest this can still preserve or improve performance. Another is Multiple Hop, where non-consecutive layers are skipped across the network for potentially larger computational savings. However, this introduces a much larger search space, as the number of layer combinations grows rapidly and layer interactions make identifying an optimal strategy increasingly difficult. Future work could explore heuristics, learned policies, or layer importance estimation to reduce this search complexity while preserving efficiency gains.

6 Conclusion

In this study, we investigated Single Hop, a weight-preserving inference strategy that skips a single transformer layer to improve computational efficiency. Experimental results across multiple model architectures, downstream tasks, and ten languages show that the impact of removing a single transformer layer is highly context-dependent. Notably, our analysis reveals a strong model-language affinity effect: while performance gains are substantial for lower-resource and non-native languages, highly optimized native languages exhibit lower improvement. Supported by robust statistical significance across thousands of evaluations, our findings suggest that not all transformer layers contribute equally to downstream tasks. Ultimately, this work provides strong evidence that language models can be executed more efficiently during inference, motivating future research into dynamic, context-aware, and multi-layer skipping strategies.

Limitations

This study has several limitations. First, the experiments focus only on skipping a single layer during inference, and therefore the findings may not directly generalize to more complex settings involving multiple skipped layers. Second, although experiments were conducted across different model architectures, tasks, and languages, the evaluated models are still limited in scale and diversity compared to the broader ecosystem of language models. Finally, while consistent performance improvements were observed, the underlying reasons for why certain layers become beneficial to skip remain unclear. Further investigation into the internal mechanisms and representational roles of different layers is required to better understand and generalize the observed behavior.

Ethics Statement

AI assistance was used during the preparation and writing of this manuscript, including support for language refinement, editing, and improving the clarity of presentation. The use of AI tools was limited to assisting the writing process and did not replace the authors' scientific judgment or decision-making. All research activities, including study design, implementation, experiments, data analysis, interpretation of results, and final conclusions, were conducted and verified by the authors.

Since our approach only intervenes during the inference stage without modifying model parameters or training data, the potential ethical risks are minimal. The proposed method does not introduce changes to the original model weights, require additional data collection, or involve the generation of synthetic training samples. Furthermore, the approach is intended to improve understanding of layer contributions and model behavior rather than manipulate outputs for harmful purposes. The authors acknowledge that language models may still inherit biases and limitations from their underlying pre-trained models, and any findings reported in this work should be interpreted within those existing constraints.

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A Evaluated Models

Table 4 presents the list of evaluated language models used in this study, including the model provider, number of parameters, number of layers, and release date. For models capable of thinking mode, we do not activate this feature.

B Tasks and Languages

Table 5 presents the list of datasets used in this study, including the corresponding language, evaluated capability, and task type.

C Significance Test Tables

We present three analysis tracks because the data have two distinct structures: MMLU is genuinely multilingual (10 languages $\times$ 57 sub-tasks $\times$ 9 models), whereas the remaining benchmarks are English-only capability probes. All p -values in this appendix are computed on per-cell maxima over the K candidate layers. Combine them would either inflate the English per-language slice or obscure the capability-specific regressions that the Logical & Math track reveals. Pooled across all 5454 paired cells the Wilcoxon test rejects H_0 with $W = 13405401$, $p_{\text{raw}} < 10^{-4}$ and $p_K < 10^{-4}$ (median $\Delta = +2.80\%$, $d_z = +0.893$, $r_{rb} = +0.942$).

Multilingual track (MMLU). Across $N = 5130$ multilingual cells, Δ has median $+2.89\%$ (IQR $[+1.31, +5.00]$) with 91.9% of cells improving. Wilcoxon $W = 12005166$, $p_{\text{raw}} < 10^{-4}$, $p_K(36) < 10^{-4}$; $d_z = +0.997$, $r_{rb} = +0.967$. All ten languages and all four MMLU sub-categories show a positive median Δ that survives Holm $\circ$ K at $\alpha = 0.05$ (Appendix Tables 11, 7). Improvement is not uniform across Joshi resource tiers (Appendix Table 8): the largest median gain appears on sw (Tier 2), $+3.83\%$. Of the 9 models, meta-llama/Llama-3.2-1B-Instruct is the weakest case (rank-biserial $+0.718$, 72.3% of cells improved).

Logical & Math track. Across $N = 288$ cells the track pools 27 BBH-CoT sub-tasks, GSM8K, HellaSwag and three ANLI rounds. Median $\Delta = +2.40\%$, 69.4% of cells improve; Wilcoxon $W = 30704$, $p_K < 10^{-4}$ ($d_z = +0.424$, $r_{rb} = +0.595$). The pooled result hides a qualitative split: BBH-CoT and ANLI improve robustly, but Single-Hop degrades GSM8K and HellaSwag across all 9 models (rank-biserial -1.0 for both). Within BBH-CoT, the regressions concentrate on a minority

of the 27 sub-tasks (*multistep_arithmetic_two* and *web_of_lies*; Appendix Table 9). Tasks that survive Single-Hop tolerate the loss of one middle-layer computation; the tasks that fail are those that require precise sequential arithmetic or symbolic state tracking — suggesting these computations are concentrated in, not redundant across, the intermediate layers.

Other capabilities. HumanEval (code), IfEval (instruction following), SQuADv2 (reading comprehension) and ToxiGen (safety) each contribute $n = 9$ paired cells and are individually underpowered; their effect sizes are reported descriptively (Appendix Table 10). ToxiGen shows the largest single-task gain (median $+7\%$, all 9 models improve); HumanEval is the most fragile (worst-case -19.5% on meta-llama/Llama-3.2-1B-Instruct, consistent with the Logical & Math regression pattern).

Table 4: List of evaluated language models.

Model Name	Company	# Parameters	# Layers	Release Date
Gemma 3 270M Instruct (Team et al., 2025)	Google	270M	18	March 2025
Gemma 3 1B Instruct (Team et al., 2025)	Google	1B	26	March 2025
Llama 3.2 1B Instruct (Grattafiori et al., 2024)	Meta	1.23B	16	September 2024
Llama 3.2 3B Instruct (Grattafiori et al., 2024)	Meta	3.2B	28	September 2024
OLMo 2 1B Instruct (OLMo et al., 2025)	AllenAI	1B	16	April 2025
OLMo 2 7B Instruct (OLMo et al., 2025)	AllenAI	7B	32	November 2024
Qwen3 1.7B (Yang et al., 2025)	Alibaba Cloud	1.7B	28	May 2025
Qwen3 4B (Yang et al., 2025)	Alibaba Cloud	4B	36	May 2025
Qwen3 8B (Yang et al., 2025)	Alibaba Cloud	8.2B	36	May 2025

Table 5: List of benchmark datasets and their characteristics.

Dataset	Language	Capability	Task Type
GSM8K (Cobbe et al., 2021)	English	Mathematical Reasoning	Generation
HumanEval (Chen et al., 2021)	English	Code Generation	Generation
IFEval (Zhou et al., 2023)	English	Instruction Following	Generation
SQuADv2 (Rajpurkar et al., 2018)	English	Reading Comprehension	Generation
ToxiGen (Hartvigsen et al., 2022)	English	Safety	Multiple Choice
HellaSwag (Zellers et al., 2019)	English	Common Sense Reasoning	Multiple Choice
Global MMLU (Singh et al., 2025)	English, Indonesian, French, Korean, Japanese, Chinese, Swahili, Arabic, Russian, Hindi	Multilingual, Domain-Specific NLU	Multiple Choice
BBH (Kazemi et al., 2025)	English	Logical Reasoning	Multiple Choice
ANLI (Nie et al., 2020)	English	Natural Language Inference	Multiple Choice

Table 6: Headline pooled results: the global Wilcoxon signed-rank test alongside the three track-pooled tests. Track sample sizes partition the global N ($5130 + 288 + 36 = 5454$). The four rows are nested views of the same data rather than a multiple-testing family, so the $p_{\text{Holm} \circ K}$ column equals p_K by construction (no family-wise correction is applied).

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
Global	5454	+2.80	90.6	13405401	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.942
MMLU	5130	+2.89	91.9	12005166	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.967
pooled Logic/Math	288	+2.40	69.4	30704	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.595
pooled Other pooled	36	+2.44	72.2	469	0.0163	0.5871	0.5871	+0.408

Table 7: MMLU per sub-category (Humanities, Social Sciences, STEM, Other).

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
MMLU-Humanities	1170	+2.23	89.8	605882	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.944
MMLU-Other	1170	+2.68	91.1	619453	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.960
MMLU-STEM	1710	+3.70	94.4	1377480	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.984
MMLU-Social Sciences	1080	+2.39	91.1	525801	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.961

Table 8: MMLU per Joshi resource tier (5=high, 2=low).

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
Tier 2	513	+3.83	94.9	126624	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.982
Tier 3	513	+2.64	93.0	120621	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.965
Tier 4	1539	+3.00	92.7	1085878	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.979
Tier 5	2565	+2.49	90.6	2957679	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.954

Table 9: BBH-CoT per sub-task. Holm over the 27 sub-tasks. Negative r_{rb} marks a sub-task where Single-Hop hurts.

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
boolean_expressions	9	+2.40	88.9	39	0.0254	0.9141	1.0000	+0.733
causal_judgement	9	+3.74	88.9	40	0.0234	0.8438	1.0000	+0.756
date_understanding	9	+2.00	66.7	30	0.1934	1.0000	1.0000	+0.356
disambiguation_qa	9	+2.80	55.6	35	0.0781	1.0000	1.0000	+0.556
dyck_languages	9	+1.20	55.6	32	0.1328	1.0000	1.0000	+0.444
formal_fallacies	9	+11.60	88.9	44	0.0039	0.1406	1.0000	+0.956
geometric_shapes	9	+6.40	77.8	42	0.0098	0.3516	1.0000	+0.867
hyperbaton	9	+3.60	77.8	35	0.0078	0.2812	1.0000	+0.944
logical_deduction_five_objects	9	+2.40	77.8	36	0.0527	1.0000	1.0000	+0.622
logical_deduction_seven_objects	9	+2.80	77.8	42	0.0117	0.4219	1.0000	+0.844
logical_deduction_three_objects	9	+1.60	77.8	36	0.0488	1.0000	1.0000	+0.622
movie_recommendation	9	+5.60	77.8	30	0.2070	1.0000	1.0000	+0.333
multistep_arithmetic_two	9	-2.00	33.3	14	0.7109	1.0000	1.0000	-0.222
navigate	9	+3.20	88.9	42	0.0098	0.3516	1.0000	+0.867
object_counting	9	+2.40	66.7	22	0.5078	1.0000	1.0000	+0.000
penguins_in_a_table	9	+2.74	55.6	30	0.2129	1.0000	1.0000	+0.333
reasoning_about_colored_objects	9	+2.80	77.8	40	0.0156	0.5625	1.0000	+0.800
ruin_names	9	+5.20	77.8	28	0.0938	1.0000	1.0000	+0.556
salient_translation_error_detection	9	+3.20	77.8	40	0.0156	0.5625	1.0000	+0.800
snarks	9	+3.93	88.9	44	0.0039	0.1406	1.0000	+0.956
sports_understanding	9	+2.40	88.9	42	0.0117	0.4219	1.0000	+0.844
temporal_sequences	9	+5.20	77.8	35	0.0078	0.2812	1.0000	+0.944
tracking_shuffled_objects_five_objects	9	+2.80	88.9	42	0.0117	0.4219	1.0000	+0.844
tracking_shuffled_objects_seven_objects	9	+4.00	88.9	43	0.0078	0.2812	1.0000	+0.911
tracking_shuffled_objects_three_objects	9	+2.80	66.7	24	0.0469	1.0000	1.0000	+0.750
web_of_lies	9	+0.00	44.4	10	0.5781	1.0000	1.0000	-0.048
word_sorting	9	+0.40	55.6	20	0.3984	1.0000	1.0000	+0.139

Table 10: Track 3 (Other capabilities) per task. All $n = 9$; descriptive only.

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
HumanEval	9	+0.61	55.6	20	0.6250	1.0000	1.0000	-0.111
IfEval	9	+1.66	55.6	24	0.4551	1.0000	1.0000	+0.067
SQuADv2	9	+1.39	77.8	35	0.0820	1.0000	1.0000	+0.556
ToxiGen	9	+6.91	100.0	45	0.0020	0.0703	0.2812	+1.000

Table 11: MMLU per language. Holm correction over the ten language tests.

Slice	n	Med. Δ (%)	% > 0	W	p_{raw}	p_K	$p_{\text{Holm} \circ K}$	r_{rb}
ar	513	+2.94	91.8	119482	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.978
en	513	+2.11	89.3	118451	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.922
fr	513	+2.53	90.8	118371	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.960
hi	513	+3.29	94.0	123858	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.978
id	513	+2.64	93.0	120621	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.965
ja	513	+2.96	93.2	123070	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.973
ko	513	+3.00	92.4	119174	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.981
ru	513	+2.63	90.3	117333	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.943
sw	513	+3.83	94.9	126624	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.982
zh	513	+2.29	89.7	115019	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	+0.968